

ActInf Textbook Group ~ Cohort 2 ~ Meeting 22

(Chapter 10, part 1)

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all right greetings thanks for joining all it's March 29th 2023 we're in cohort II in our first discussion on chapter 10. so let's head over to chapter 10. and um begin with any comments really about anything within the active area several were voiced before we beat God so perhaps Terry feel free to begin with um what you wanted to point towards and let's just see where that goes thank you uh in in essence my my question is about the concept of a [Music] um a system of volitional Consciousness behind that sits behind the uh nodal active inference model and creating uh I suppose uh making decisions about about the Strategic direction of the organism and um I I find just repeatedly within this chapter there seems to be this reference to you you know a a conscious entity behind the algorithm and my sense from to trying to understand this material is that it's almost like um in meditation where you know the the the the the experiences the experience that that that almost there is no I behind the experience and that it just jars with me when we're presented with a almost complete uh active inference model or that that where where our our actions in the world are actually experiments to gain information about the external state to um up up ultimately update the model and what Consciousness is as you said at the start is is a difficult concept but um it seems strange to me within the context of this model that we refer back to some sort of uh underpinning strategic conscious experiential driver of action when that seems out with this model thank you are unnecessary for this model Michael Raleigh want to give any first thoughts on this yeah Ollie go for it yeah uh well actually I don't think uh they necessarily uh somehow uh I mean the active inference framework doesn't require any conscious entity in the sense of our anthropic sense of this word Consciousness because what FTP actually tries to model is a kind of uh quote-unquote as if epistemology still by uh as if epistemology it doesn't mean that everything that uh can can be modeled with fpp formulation uh necessarily um has anthropic uh this sense of anthropic Consciousness but it's a kind of a kind of uh self-referential way of talking about uh the perspective of being modeled so I'm sure you've encountered this word self-evidencing

throughout this book which refers to the same concept of as if epistemology so it's or um one of I'm sure Daniel's favorite figures um in the book figure yeah 9.1 also captures this sense of uh perspectival uh view view of uh kind of the nature of this modeling is not that we have this sense of uh omniscience that that can be applied uh to the environment or that can be uh reasoned through uh with uh I mean reason through and uh that can also be applied to our sensorium or perceptorium but it's like we have this uh model inside a kind of apparatus that can see Its Behavior as if uh following through a kind of trajectory in the probability density of space so by volition uh that you're referring to it's my understanding that it's also another term that they've also used that's related to this self-evidencing is uh they is the object or the agent tries to infer its environment uh through a kind of uh what was this term in it was um something like indirect pers no it wasn't Daniel can you help me find the term for uh the indirect perception of the specific term that's used in the literature to refer to indirect perception uh sorry just a second I don't exactly remember okay I'll find it and I'll get back to you in just one second okay and I mean I I I I stop me if I'm going down some unnecessary Rabbit Hole um uh but it's almost as if uh this system active inference represents a model by which one could uh create a psychological zombie um in in that you know by active inference without any yeah extensive internal self one could behave in the world uh by um you know minimizing uh variational free energy in the way that we behave in the world and Daniel then it seemed to him one of his books sort of talk about that um uh the RR you know that we have active won't rather than active um uh that that than rather than free will you know and um I I suppose the the implication of that is that we um that that our conscious reality is is like a a story We Tell ourselves almost uh shortly after the event and yet in this book there is this reference to uh uh a conscious will within the within the framework or is certainly within that last chapter uh chapter um uh five that we're isn't it chapter five we're on today so I I don't I I I I mean I don't have this worked out of my head but um I'm just interested it's entirely possible that um one could say active inference isn't about what it it doesn't speak to Consciousness Consciousness is another thing active influence just off uh gives another is a potential model that uh drives um decision making within the organism almost and that's okay um but I'm just in I I'm just interested in other people's thoughts on this it's it's it's great topics so just a few notes broadly there are perspectives on Consciousness well first I'll just note Consciousness appears to be only used once in the book so while it is true that um it's an area of interest and was the live stream with uh vanja Visa this morning I mean and the authors all explore elsewhere but any interpretation of consciousness I I don't even think it's within the it's

not a pillar of the framework it's something that is like inferred towards so I think that will always leave a fundamental um uncertainty in pluralism and then we can think about at least a gradient of um accounts of Consciousness in some accounts Consciousness is um after the fact in some way that it could still be anticipatory from a predictive processing sense but it does not have a causal power it's it's a narrative told to um let's say minimize variational free energy to make sense of memories and so that kind of narrative expositional self as epiphenomena and experience without causal impact and then that is of course contrasted with causal accounts of consciousness and that's where whether you come from some philosophical tradition or certain folks psychological Traditions um which both can arise during scientific writing as well things like emotions or awareness or even attention are actually described as causally Upstream of other phenomena and that puts one of course squarely in the kind of Cartesian dualism area of philosophy like how does something that is not physical influence something that is physical um so the the the the dominant mode to me on that question is Consciousness is something we can model about and around with active inference but no perspective on Consciousness seems fundamental to using statistics in this way and so I expect and already find confirmatory evidence that some people take a more Epi phenomenal approach to Consciousness and some people taking a more causal approach for example treating it like a hidden latent factor that does have a causal impact in a causal Bayesian sense and then also just recognizing yeah we don't know exactly what it is that's how we do active modeling we propose latent clauses and those become our generative model so in some ways there may be more similarities between these two than their hyperbolic forms suggest and in either case I feel like it's a Continuum that's accommodated methodologically by what Ali referred to as the as if and that I think connects to Daniel dennett's intentional stance like I'm going to play this chess computer like it's trying to beat me if it's not if it's to try then I then I want to win so if it wants me to win two then that's going to be great but I'm gonna hedge my bets and play like it's trying to win and I think that kind of um intentional stance is not just warranted it's actually supported by these methodological tools Terry or Michael well in some ways and we're getting way off topic and then I will be quiet oh no this is we're technically in chapter 10. so we're we're totally on just recap and meme status oh okay but we but it's like but you would say that wouldn't you you know I I I if Consciousness isn't after the fact um uh Story We Tell ourselves um you know it's at the model if the world is probably more likely that one finds oneself in front of a computer play playing chess uh because of the you know all of the previous events that have occurred during your your life and that you are now playing

chess so the story you tell yourself is that I'm going to play to win but actually that it wasn't the it wasn't the conscious decision that got you there if Consciousness is and after the fact uh you know hallucination or or story that we that we create to justify the things that are happening in the external in the on the One external State you know because the computer is in the unknown external State it's it's a it's something that we have internally created um uh as our model based on the information we're getting across our Markov blanket am I getting too mad I'll give a short note and then Ali so um one answer from a biological perspective is especially over evolutionary time but also within a generation you will only repeatedly observe things that are fighting to live and persisting amidst challenging and adversarial settings so if we just say oh well just the person in front of the computer what would be subconsciously leading them to want to win but if we say we're in a chess tournament series and only people who have won previous chess tournaments defined by winning in this arbitrary way only they are persisting in this tournament well then over time you will come to find that those agents are acting as if they want to win at chess now then there's the debate well what if they're just acting like they want to win at chess but they don't really want to win they're they're chess zombies they're just acting like it and then that's the pluralism some people look at that chess tournament sequence and they go yeah I actually have like reasons to think that people don't have experience or intentions or that Consciousness is just a narrative told after the fact and so win and the other person tells us extremely intimate causal story about how the the motivation of the person drives their action selection but in the end they're unified by this as if epistemology which is to say the intentional stance Ali uh yeah also this uh kind of as if comes from at least partially comes from uh kantian transcendental idealism so for uh Concepts do not apply to reality in the form in which we can only think them but it is also urgently required for any theoretical practice to think and investigate objects uh with the help of the concepts as if they were given or determined um in a specific way so this kind of as if epistemology is it you see uh Concepts have to um acknowledge that a reality is presuppose independently of the choice of uh its semantics or its meanings on the other hand in their self-reference and self similarity uh they work with this heuristic of as if so that uh it can be constructed in this way as if reality and concept would actually coincide with each other but some recent philosophers such as Karen broad attempts to overcome this kind of as if epistemology uh for example in her agent of realism agential realism and by means of construction of interactive discursive in which both epistemology and and ontology can be seen as the very same thing and we wouldn't be required to draw a boundary between the

perception on the one hand and the thing that we're trying to percept or model on the other hand so through this uh intraction not interaction but interaction uh or constructive interaction uh they constitute each other so as they give rise to their very existence and uh also I think in some View FTP is perfectly compatible with this more loose sense of onto epistemology because one of the requirements of the fpp is if things exist how they should behave right so it's somehow presupposes this kind of interactive discourse by basically eliminating the boundary between the thing that we're trying to model and um I mean the the model the model or the map so that's basically uh the argument behind that map territory fallacy fallacy uh so uh yeah this is a bit uh more complicated uh in in this sense that uh if we want to get to the bottom of this kind of epistemology uh we will hit a dead end if we just if we just uh go through this more traditional epistemology of uh of Western epistemology of uh distinguishing between the thing that what we're trying to perceive and the perception itself uh but as I said fep somehow tries to overcome this problem in the first place hmm that's that's beautifully um uh described and nicely closed down and that the thing that I was thinking initially was um the assumption that in the external State there's actually AHS competition happening um is assuming almost this uh car you know uh that that that are um internal state representation is an accurate or a true likeness of what the external status which I think Holly was kind of saying but um I I my my sense is that it clearly can't be because all of our um all of our all of our qualia are Endeavors that the the the the the the the the central nervous system uh makes to interpret um signals that are generated without the qualia that we apply to them yeah you know these are these are great one sure thought then Michael um yeah the internal state does not in the model does not have to have the same structure as the external state so like often we'll talk about well the external state in the model could be a continuous temperature and the internal State could just be maybe too hot just right or too cold so they can have different structure and that reflects that they don't even have to be of the same type even within the model let alone the territory however by the cybernetic principles like good regulator theorem again eventually you can always just create an edge case where the agent has no um mapping between their internal cognitive Dynamics and the external regularities like it's just playing the game randomly with respect to the environmental variability and in a research or an education setting that may be pedagogical but when we're talking about real measurements that we made even of like the the pen on my desk it is resisting dissipation over some spatial temporal time scale and that is actually the surprising implication of if things exist how they should behave is like well I am repeatedly measuring this pen so now what can I say about that and it turns out

you can actually say a lot if you bake into your understanding that you are observing something repeatedly existing and that's exactly what we see in figure 9.1 and so we we have a a Unity of the scenario in our model via articulating the partitions on a base graph so that means they're still totally up in the air map and territory and all of that but within our model we're we're approaching that in a way that sets the stage for people to then take further perspectives about deeper questions Michael yeah I I think that these questions may be uh somehow relate to this Tale of Two density papers um great you mentioned last time so and I my understanding of this paper is that they um try to argue that there's no representation in the mind but the mind is a controlled system and then they talk about two different distributions that I so so uh so so one distribution is the recognition distribution and the other one is the generative system if I understood correctly but I didn't really understand how this distribution how this this distribution comes into being through the um generative system so that equal one with an enacted so the recognition system is the an embodied system and the generative system is the enacted system and my understanding is that this is all that there is so for instance they say here the on page 235 the generative model manifests as a control system that uses exploitable structural similarities encoded in the internal states of the organism so instead of having representations they talk about controlled States and to understand this correct so how can that be um a distribution or I I didn't really I didn't really unders understand where these two distributions are great all great questions so I'll try to address some so um this question of exploitable similarity is very much like the good regulator theorem in cybernetics so it doesn't mean that the temperature in the room let's just say that it's physiologically relevant so there have to be some sensors and some processing apparatus and also action apparatus but at the very least just for cognition a sensory and processing apparatus that is able to like measure temperature it can be a proxy of it doesn't need to be a strict isomorphism they say it's weaker than a strict isomorphism but it actually has to have some sort of causal relevance now if you're like well I'm in a simulation where the temperature is fluctuating but it has no survival value then it's like okay that's like saying you know cosmic rays are going through us they don't have survival value and so they're not part of the cognitive model in this setting um but um this points to this idea that there is information in the sensory apparatus about action it's actionable intelligence with respect to persistence why persistence well we could say in pursuit of reward but then we would need to ask how systems persist why persistence because we're at least repeatedly measuring the system so again with the as if epistemology it's as if in order to persist amidst dissipative and adversarial settings it's as if

there's an exploitable information content in the perceptual Umwelt that enables that kind of thing to persist that's like the tautology that also goes beyond a tautology now to the specifics of the tale of two densities um this also might have been the first dot zero in in six I'm I'm not sure in Bayesian statistical analysis models can be run in two different directions this could be like a hyper prior and and um here's the hyper priorities on the right and the data are on the left so it's like you can specify the hyper prior or the hidden State distribution and you can emit pseudo data that's called generative or you can have data and then parameterize hidden States that's the recognition model so this could be the actual measurements of children's height in the classroom here's our generative statistical model gaussian family mean and variance so given a mean and a variance estimator two numbers metaphysical statistical we can generate infinite class size of synthetic height data or from one height observation or we can update our generative model of the world okay then more broadly we're interested in this relationship between the world and the agent agent in the niche so perception and how that updates the agent's learning and uh and perception and then how the agent acts on the world and what they argue in this paper is that the recognition model in a Bayesian statistical setting so going from measurements which might be noisy and the the height in centimeters isn't like the real height of the person the number on the scale isn't the weight the number on thermometer is not the temperature like it's data this is why the empirical perspective on behavioral modeling is so essential because when we go on this Bayesian side it's really easy to talk about how data are recognized and update hyper parameters and then hyper parameters can be in the business of continually generating unfolding predictions we have a noisy camera with a blind spot we're updating our generative model of vision and then we're going to be in this case just creating more synthetic Vision data in an activism and all of the associated E's embodiment Etc et cetera et cetera et cetera they are focused on the materiality of the engagement of the agent with the world in this paper Tale of Two densities they draw these two together to argue in the context of what we'll just point to in figure 4.3 with the a matrix here two there's a generative production of synthetic observations you could use it that way that's the generative direction of a or you could use a in its recognition capacity to update hidden States from observations so even without getting into the generativity of action which is what Indian activists setting what they're most interested in is the actual causal consequences of action just within a figure 4.3 hidden State inference we can think about a tale of two densities and about how Bayesian statistics helps reconcile this Tale of Two densities because the same generative model which is just what we call whatever base graph you construct the generative model can be used

equally flexibly to do inference about any part of it like maybe um you know the hidden States but you're trying to determine the prior so these graphs are agnostic as to what you know and what you want to infer now it turns out that for real settings that we find in the behavioral lab and out in the fields we get observations we don't have access to Hidden States but we want to infer them we don't even think A or B matrices or D matrices exist in the world we purely want to infer them as statistical parameters and then we want to know about action and in some settings we have the um organism's trace of actions and we're trying to infer their cognitive parameters that's the figure 9.1 setting we saw their actions it was data for us and we're looking to parameterize a cognitive model for them another setting is we don't know what action will be taken because we're the ones taking it and that is like a prospective application of these types of models to take ongoing streams of observations and then perform action selection in the moment but these Frameworks are enormously General with whether they're used for online or offline inference and like what you actually have access to measuring and how so little long but the tailor two is about how the Bayesian epistemology and structuring of the relationship between observations and inferences about observations maps in the inactivist setting to some aspects of how we think about perception cognition and action in the niche and their paper positions that within the debates around representation because representation are although there's different perspectives Etc postulated to be like the on agent aspects that have semantics about the world yeah and they say the agent just controls the world through their own actions that's my understanding yeah so then then the two densities are on the incoming side and in the out outcoming side okay that's the idea the density is not the the constitution of the states of the internal States but the observable actions we may be stretching the tale of two densities making more of a um an um a set of novels around densities because it's just ultimately a cultural reference the the technical point is that these variables are statistical distributions and hence they are densities their statistical densities over some support the two densities that we might be interested in or amongst two of the ones that we're interested in making a unified account of are the perceptual density which is about how data map on to Hidden States like given retinal data is that a car that's a recognition model and then there's kind of like two kinds of generative densities you could run that car recognition model backwards say given the label of a car generate me cars stable diffusion generative AI but then there's also this action as a generative step that brings in the actionable and cybernetic um constraints which is why they focus on the exploitable but in general Bayesian models you can do this kind of observable hidden State inference about anything the

organism Bayesian brain or if we choose to use Bayesian statistics it's going to be doing something like the world has a real temperature which is like molecules moving at different speeds perception occurs through sensors which transform whatever it is the molecules are doing into something totally totally different can I just uh make a little uh point about that that um that I I remember watching you know Richard Feynman just a short video where he was describing temperature is actually infrared radiation which is part of the electromagnetic spectrum so on one part of that Spectrum we see light and on the other we experience temperature so temperatures actually a proxy for um something else that you know what we the the mechanism that we have to perceive temperature is then a proxy for the amount of energy within that for example room so it it's not our response to the infrared information is is a response that's not based on infra necessarily that infrared is going to damage us it's that there's too much energy in this room and the tool that I have uh to assess that energy in the room is actually monitoring the infrared uh radiation which I can't see but I can feel any comment on that yeah like by analogy um how about tasting different foods and there might be a toxin like we can't directly tell that something is going to be contaminated but over time successful entities in an unchanging niche now that's not every case but just with that simple case at first successful entities in an unchanging Niche will come to exploit similarities which is to say Mutual informations between the recognition model and the actionability and so they don't need to be directly measuring or even internally representing this is going to make me sick or this is the wavelength of light um all any and all informative cues of sensors are going to be aligned towards adaptive action that's the synthesis of signal processing and control theory in active inference and it turns out that by by taking a first principled approach to the synthesis of those two fields again just coarsely we have new insights on the question of in cognitive science yes but taste is a very interesting phenomenon and that we have sweet salts are bitter and yet there are I think there are trillions of different tastes that we can that we can uh that we can actually uh create or are I think smell is probably associated with taste as well so you know from a limited amount of sensory information we can create a vast array of experience yeah I mean with the um the Jolly Ranchers I don't know just a candy had in childhood like they literally taste the same if you don't see the color but if you do see the color even if you try to forget they will taste different so yeah so different um that's like referred to as Sensor Fusion difference sensors can come together in the generative model and it really speaks to the prediction yeah well it really speaks to what are our what are our prayers you know another kind of mind bender on this is like the visual example there's so much abundant

um on-ramps to predictive processing like we don't notice our blind spot we don't have any blur when we do the saccade several times per second because of sensory attenuation you know color vision and Precision in the periphery okay I'm on board with a generative model of vision I'm I've kind of dispelled this sort of incoming information camera processing model of vision I really believe in the visual generative model getting updated and it's like but taste it's like Yeah by extension what does that really mean that we're generating our taste experience well then why wouldn't I just generate tastes that tastes amazing like well why what why wouldn't your generative model of vision just generate what you like to see and that's where the real organismal constraints and where appeals to our development ecology and evolution do come into play so again contrived gedanken can be proposed absent these super obvious constraints for real systems however in the context of empirical behavioral modeling it's usually sufficient to say the prior Distribution on this or the trade-off between these two different drives is inherited from history and development or even from intergenerational processes but but but isn't it that we generate a good taste going to a baker where we like to go and doing and going towards those areas where we get those kinds of rewards that we prefer I don't know why yes okay okay here we were okay chapter seven we were engaged in a policy vacillation we were not sure if we were wanting to get bread or if we want to go to Barber so now the question of the taste yes if we expect and prefer a given taste like sweet and I think this holds true for sugar but it can hold true for other things too in the free energy evaluation of different policies with expected free energy expecting observations that align with your preferences will be associated with pragmatic value so that is how preference for sweeter tastes puts its thumb on the scale of policy selection biases policy selection towards tasting sugar as a pragmatic reward but that happens amidst a relationship with epistemic value as well so it's like um epistemic value could be pursued like we need to find out how to get to the baker I mean there's there's many ways to go but yes you would parameterize a generative model with a preference let's think about this this case with sugar in the baker in those two settings that we described so in one case we're in figure 9.1 and we're we're observing people's behavior so we have their credit card history whoops and we're observing who is going to the baker or what they're ordering or what they're clicking on online and now we're going to do inference on a generative model for them it's the as if and it's like wow this person has a very strong C Vector very strong preference Vector for sugar like given Alternatives they're always going for the sweeter one and this person has a neutral preference for sugar and this person actually like like moves away from sugar as represented by our

parameterization of the C vector so that's one setting we have data on a bunch of different entities on what they chose in different sugar concentrations and then the other setting would be now step into the trajectory of one of those cognitive models and now ask what your next action should be and that's the as if a gentle or intentional stance so this is the behavioral scientists view from the outside and then that is going to be flexible with allowing us also to take views as if from the inside and just because we can take an as if View From the Inside doesn't mean that that's the territory that's literally the map territory fallacy but it means that within the limits of cognitive model we do have that opportunity uh what one of the things that strikes me is you talk about this is is the one of the interviews Carl Friston has done online he talks about you know our actions during during the day that you know we get up and we um have our breakfast and then we um you know we we go to the bus and we get go to work and we do our work but we I you know and actually most of our life is um little routines we we think that we have these independent lives where we are constantly making uh decisions about what what you know what what what do I want to eat today what where will I go but actually the person who ends up going to the baker will find that on their walk to the bus stop they pass the bakery and they get a trigger from the smell of the bakery to do the thing that you know to predict that it's it's it's just it's interesting that we don't exist but as these completely independent no matter where you live no matter who you are there are so many aspects of your life that are just so so repeatable that we actually live within a very constricted box apart from odd times when we step outside that box I I I I don't know if that's relevant to what to that picture but oh it's awesome I mean very very interesting to point to the state of exception in live stream 10 with um Mao and ramstead at all who are the other authors Mao Axel Carl and Maxwell um they talk about like strong and weak Scripts a strong script in a social setting would be like if we were literally reading from a screenplay or if we were doing some sort of Highly ritualized Behavior whereas the weaker script concept refers to like Norms normativity expectations X kind of person should act like this I'm X kind of person I would be unsurprised by doing X up what do you know I'm doing X and that kind of progressive identity capture and development of understanding what kind of thing one is and then what are the two ways you can reduce your discrepancy you can change you can learn change what you think that kind of thing is and you can act so undergrad should act this way you know I expect here's the distribution of undergrads if one's um understanding of what an undergrad was was totally flat with respect to everything there's nothing that characterizes an undergrad versus anything else just from one person's perspective well then that aspect of their identity wouldn't shape their

policy selection whereas if someone says well as a x I should be doing why that can be approached from a purely statistical deflationary perspective like I just expect myself to be doing this like or it can go even beyond that too so those are these are all very relevant and learning about the fundamentals of this kind of statistical modeling help us I hope I prefer also that it helps us actually respect the complexity and the richness of these phenomena which just simply cannot be expressed in words or math which is why we talk about the map yeah one of my wonderings about all this because I do find the maths and the uh the the pictures and the diagrams all very difficult to to to grasp in some way it is there a potential that if we really understood the model we could start uh uh use that as a system to inform our understanding of individual behaviors I mean I'm a pain doctor so I um my my my um my interest and all this is this this this truth that we can alter our you know if you are in pain if you have back pain you can alter your prediction about your back pain and actually alter your back pain and I'm just wondering if this kind of modeling could in some way give us a tool to help us to support other humans in understanding their ability that they can update their prayers you know yeah it's a great Vision I think the closest that we've explored that is um with the help of Ian tenants who has a interception um focus and this Therapeutic Alliance concept where where a lot of these kind of avenues and like it's not as simple as just wish it away any more than the generative model of vision would be like I want to see you know a unicorn in front of me so how do we reconcile that agential realism that Ali mentioned earlier like this active co-construction and negotiation and sense making from our sensorium and the ways that our narratives and the scripts and the expectations shape our inference and action choose to have a preference and to act to push things in a certain way absolutely yeah but but but I I suppose what what I'm kind of thinking is could you have a like a a ma you know a a Sam Altman created model where you run the model and that can inform how that can inform the uh understanding of an individuals how they got there and potentially offer some insights into what to do about it you know that that could this speak to a model that could be run where we could kind of do the experiment and then that supports an individual on actually going on the course of action to change their experience that that's that's kind of where I what what what really blows my mind about this and that in the current world that it has only come into existence in the past month or so this actually seems almost plausible because because of the the similar layered nodal systems that the likes of chati PT are using to protect language yes it's a great it's not active you know it's not active you could model it around a notion of active inference and that you know the the training of the there's there's so many Avenues to explore

there let's in our discussion next week let's look at um that again but absolutely with respect to figure 9.1s about us the data exists the means motive opportunity anyone who wants to can do things and that's a big question so for next week thank you though I'll stop recording so